

Summary: In this workshop you will extend the work done last week by finetuning VLMs with Low-Rank Adaptation (LoRA). You are encouraged to work in pairs to speed up experimental results; for example, one of you doing task 1 while the other works on task 2 – or both in the two tasks trying different hyperparameters. You are also encouraged to use this workshop for assignment support.

You are required to consider the following values for your choices of LoRA-based finetuning:

r	α	Adaptation strength	Computational Cost	Description
2	4	Very low	Very low	Minimal domain adaptation
4	8	Low	Low	Light domain adaptation
8	16	Moderate	Balanced	Commonly used setting
16	32	High	High	Stronger domain adaptation
32	64	Very high	Very high	Heavy domain adaptation

Task 1: Finetuning a Vision Transformer (ViT) with LoRA

There is only one program for this workshop, `python ITM_Classifier-LoRA-baselines.py`, which is an extension of its equivalent from week 5. This task has four steps:

1. Open Visual Code, as in previous workshops, and edit to make sure the following:
`MODEL_ARCHITECTURE = "ViT-LoRA"` and set your values for `self.r` and `self.alpha`.
 Save it if anything changed.
2. Run the program and install any dependencies required.
3. Try to understand the implementation of LoRA for ViT, look in particular at class `LoRALinear` and function `apply_lora_to_vit()`.
4. Record your results, compare and discuss.

Whilst your program runs, **think about how to combine LoRA with the materials of week 5**. This experiment could be a model that you could consider reporting as part of your assignment.

Task 2: Finetuning a BLIP Vision Encoder with LoRA

Repeat the steps of task 1 but this time using `MODEL_ARCHITECTURE = "BLIP-LoRA"`. Try to understand the implementation of LoRA for BLIP by looking at function `apply_lora_to_blip()` and where is invoked inside class `ITM_LoRA_Model`. Note that instead of using the full BLIP model, we are finetuning only the BLIP vision encoder. *Can you identify that in the code?*

Task 3: Assignment support

Please ask questions about the assignment – either task 1 or task 2. This is a good opportunity to get your questions answered, well-in-advance before the deadline and with sufficient materials covered.